

Name:Murali Manokar D K

24/07/2026

Reg:192324044

Code:

```
> library(corrplot)

corrplot 0.95 loaded> library(reshape2)

> library(RColorBrewer)

> library(hexbin)

> data <- read.csv("employee_salary_dataset.csv")

> numeric_data <- data %>% select(where(is.numeric))

> ggplot(data, aes(x = Monthly_Salary)) +
+   geom_histogram(binwidth = 5000, fill = "steelblue", color = "black") +
+   labs(title = "Histogram of Monthly Salary")

> ggplot(data, aes(x = Department)) +
+   geom_bar(fill = "orange") +
+   labs(title = "Bar Chart")

> ggplot(data, aes(x = Department, fill = Gender)) +
+   geom_bar(position = "dodge") +
+   labs(title = "Grouped Bar Chart")

> ggplot(data, aes(x = Department, fill = Gender)) +
+   geom_bar(position = "fill") +
+   labs(title = "Stacked Bar Chart")

> ggplot(data, aes(x = Department, y = Monthly_Salary, fill = Department)) +
+   geom_boxplot() +
+   labs(title = "Box Plot")

> ggplot(data, aes(x = Experience_Years, y = Monthly_Salary)) +
+   geom_point(size = 3, color = "blue") +
+   labs(title = "Scatter Plot")
```

```

>
> ggplot(data, aes(x = Experience_Years, y = Monthly_Salary)) +
+   geom_point() +
+   geom_smooth(method = "lm", se = FALSE, color = "red") +
+   labs(title = "Regression Plot")
`geom_smooth()` using formula = 'y ~ x'
> ggplot(data, aes(x = Experience_Years, y = Monthly_Salary)) +
+   geom_line(color = "darkgreen") +
+   labs(title = "Line Chart")
> ggplot(data, aes(x = Age, y = Monthly_Salary)) +
+   geom_area(fill = "skyblue") +
+   labs(title = "Area Chart")
> ggplot(data, aes(x = Gender, y = Monthly_Salary, fill = Gender)) +
+   geom_violin() +
+   labs(title = "Violin Plot")
> ggplot(data, aes(x = Gender, y = Monthly_Salary, fill = Gender)) +
+   geom_jitter(width = 0.2) +
+   labs(title = "Jitter Plot")
> ggplot(data, aes(x = Experience_Years, y = Monthly_Salary)) +
+   geom_hex() +
+   labs(title = "Hexbin Plot")
> ggplot(data, aes(x = Monthly_Salary, fill = Gender)) +
+   geom_density(alpha = 0.5) +
+   labs(title = "Density Plot")
> ggplot(data, aes(x = Experience_Years, y = Monthly_Salary)) +
+   geom_density_2d() +
+   labs(title = "2D Density Plot")
> ggplot(data, aes(x = Experience_Years, y = Monthly_Salary)) +

```

```

+   stat_density_2d(aes(fill = after_stat(level)), geom = "polygon") +
+   labs(title = "Contour Plot")
>

> ggplot(data, aes(x = Department, y = Monthly_Salary, fill = Department)) +
+   stat_summary(fun = mean, geom = "col") +
+   labs(title = "Column Chart")
>

> pie_data <- data %>%
+   count(Department)
>

> ggplot(pie_data, aes(x = "", y = n, fill = Department)) +
+   geom_col(width = 1) +
+   coord_polar(theta = "y") +
+   labs(title = "Pie Chart")
>

> ggplot(data, aes(x = Experience_Years, y = Monthly_Salary,
+   size = Monthly_Salary, color = Department)) +
+   geom_point(alpha = 0.7) +
+   labs(title = "Bubble Chart")
>

> corr_matrix <- cor(numeric_data)
>

> corrplot(corr_matrix,
+   method = "color",
+   type = "upper",
+   tl.col = "black")
>

> heat_data <- melt(corr_matrix)

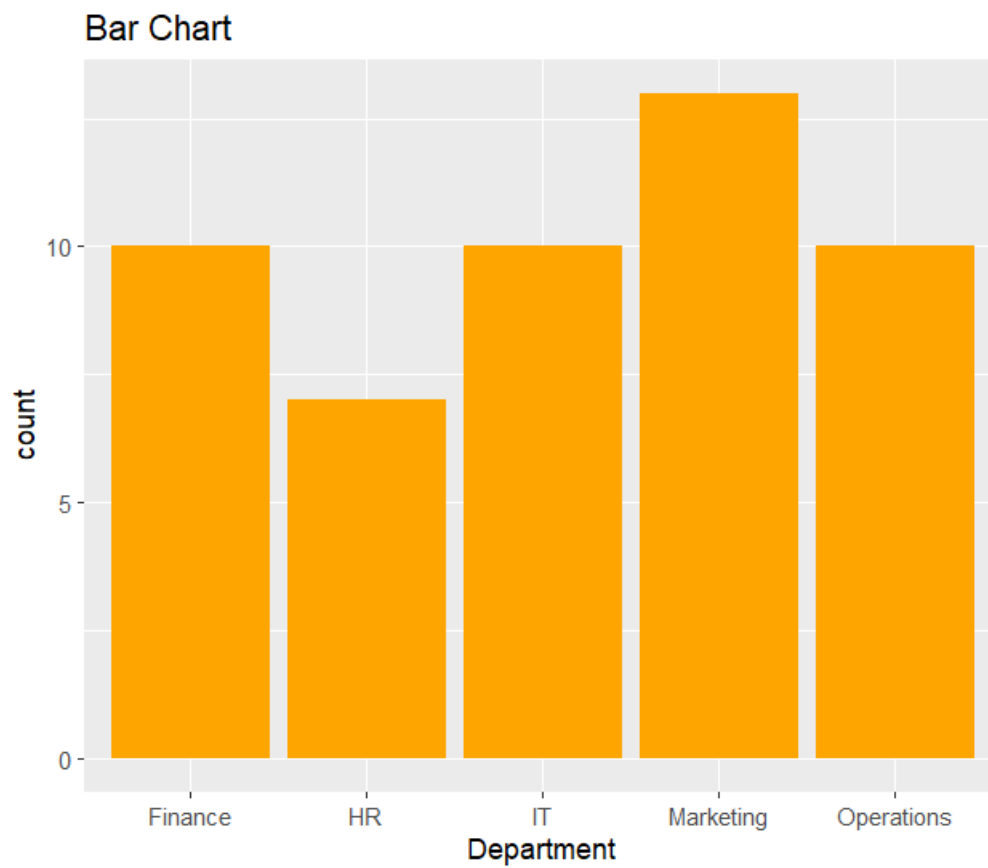
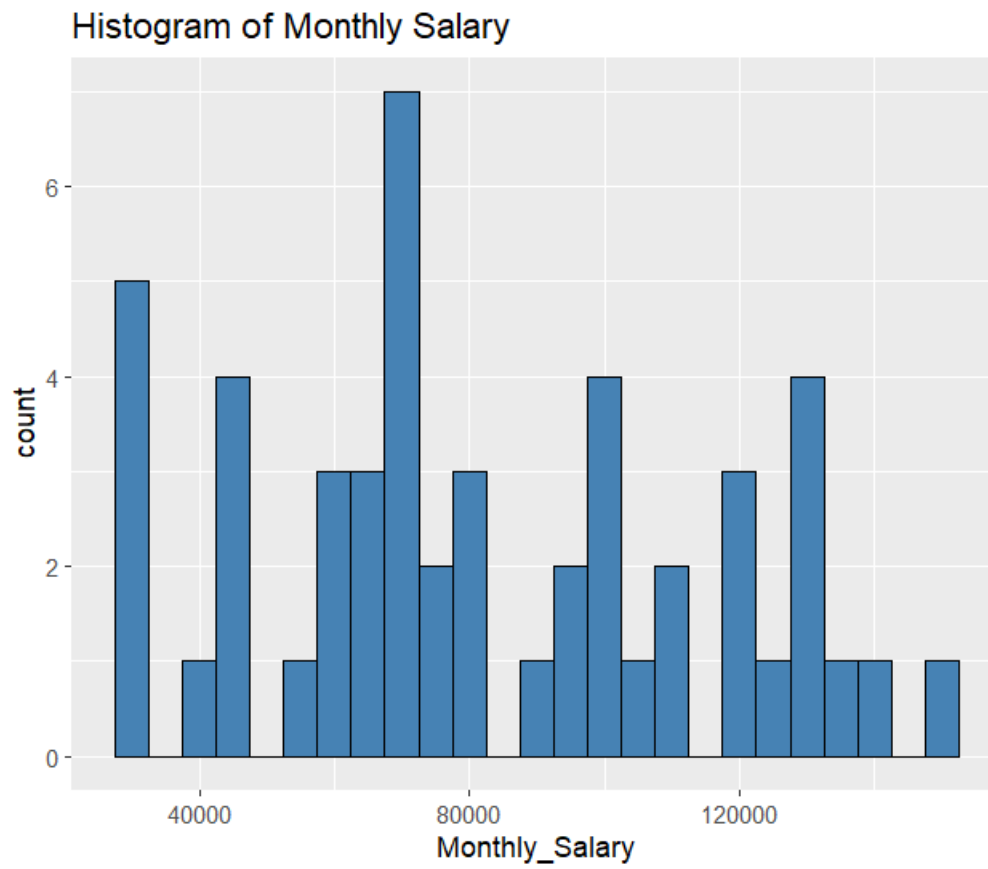
```

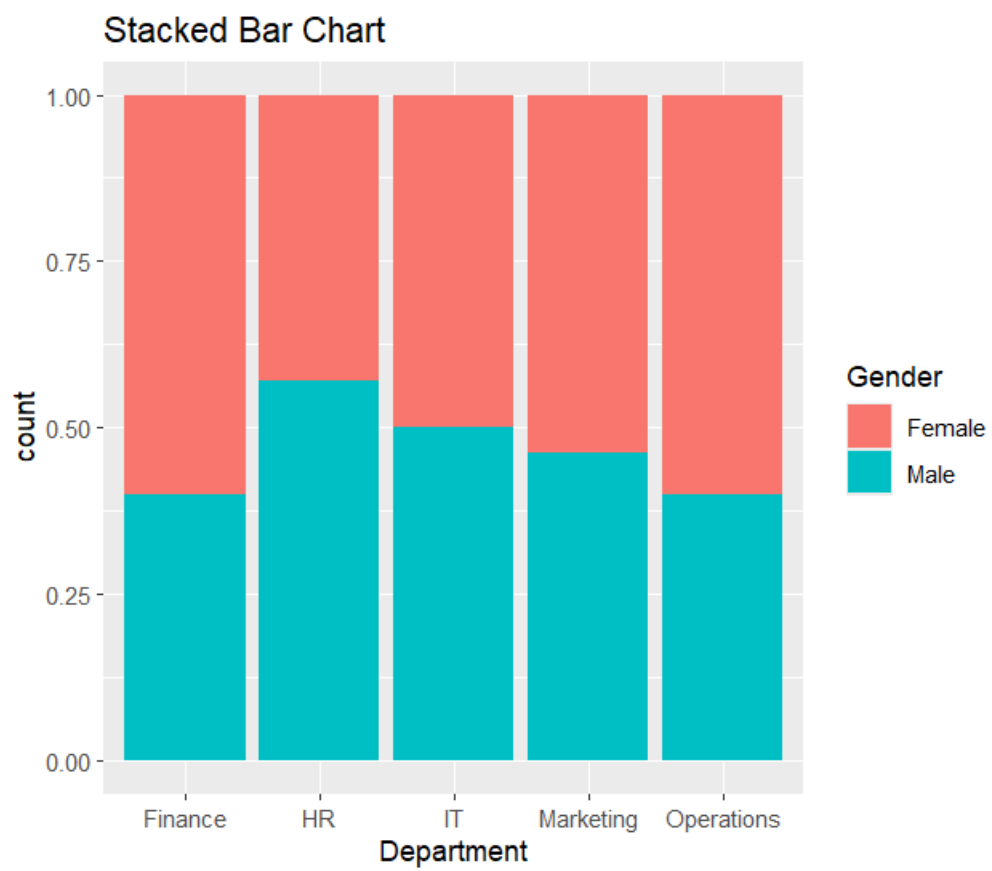
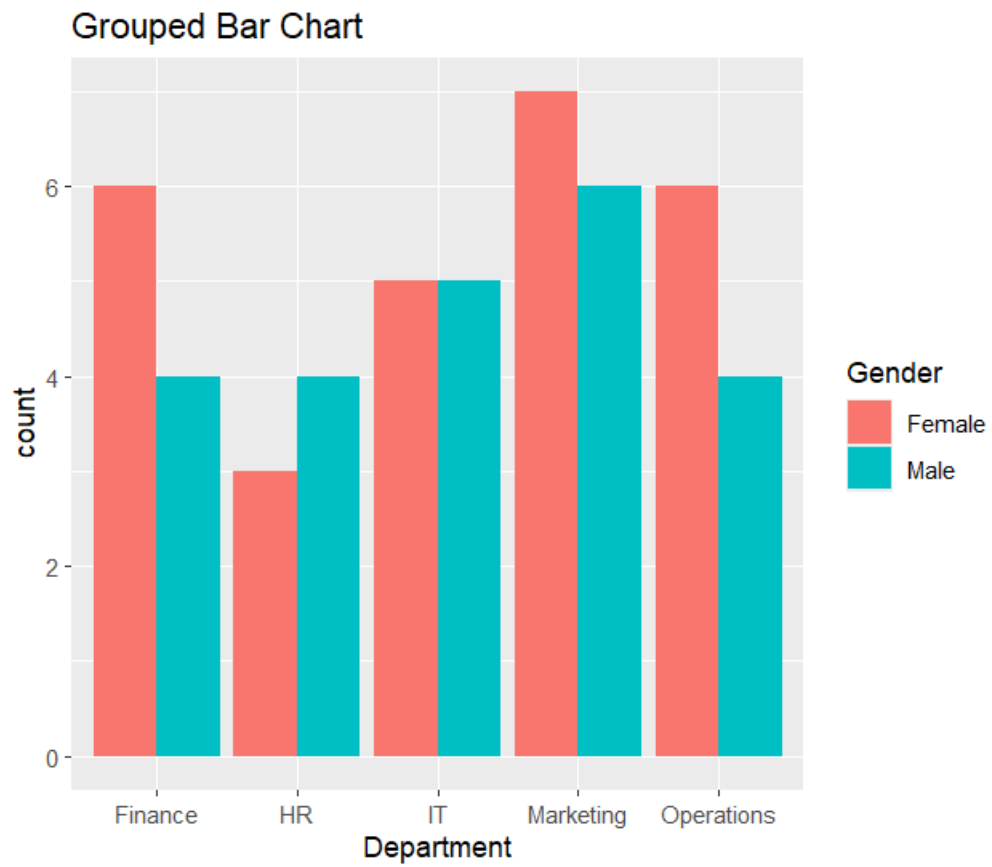
```

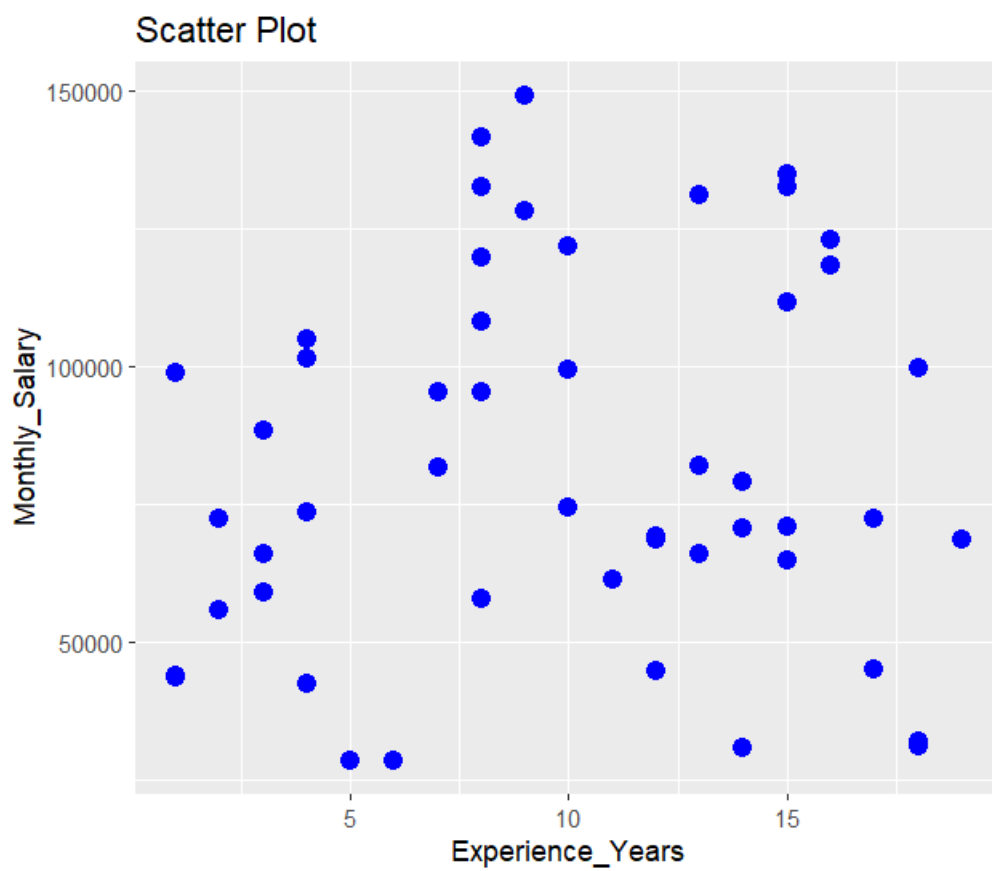
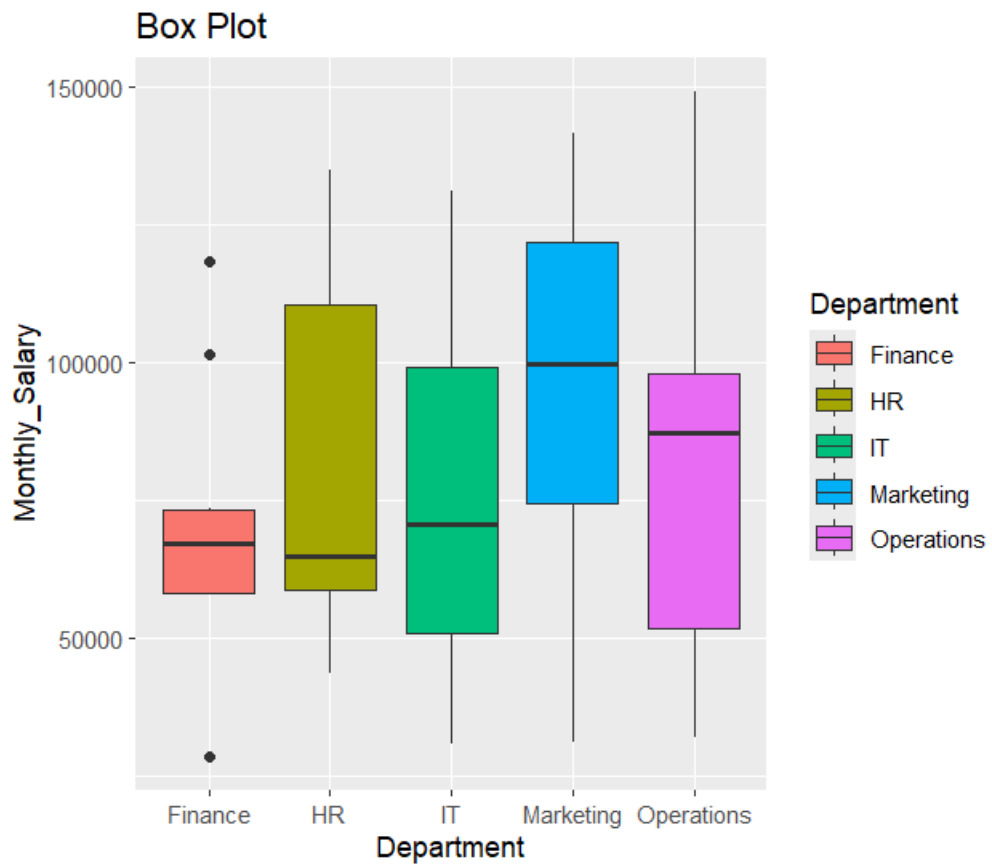
>
> ggplot(heat_data,
+   aes(x = Var1, y = Var2, fill = value)) +
+   geom_tile() +
+   scale_fill_gradient2(low = "blue",
+                         mid = "white",
+                         high = "red",
+                         midpoint = 0) +
+   labs(title = "Heatmap")
>
> ggplot(data, aes(x = Education_Level, fill = Education_Level)) +
+   geom_bar() +
+   facet_wrap(~Gender) +
+   labs(title = "Facet Bar Chart")
>
> ggplot(data,
+   aes(x = Age,
+       y = Monthly_Salary,
+       color = Department)) +
+   geom_point(size = 3) +
+   facet_wrap(~Gender) +
+   labs(title = "Faceted Scatter Plot")

```

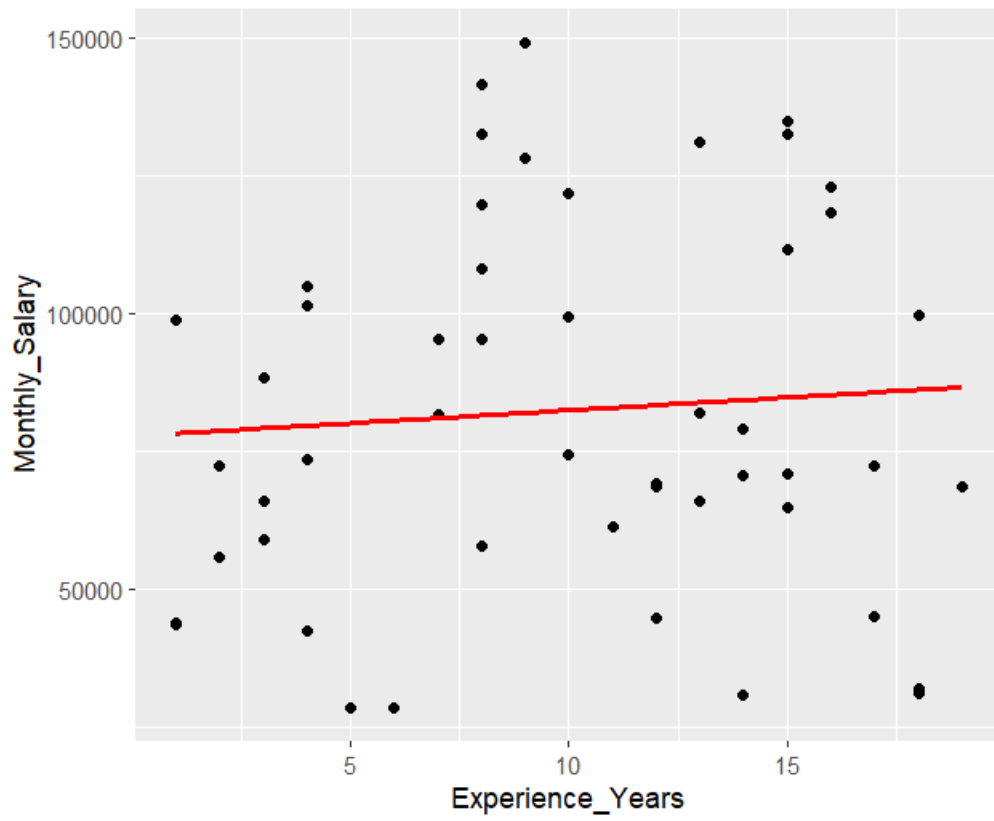
Output:



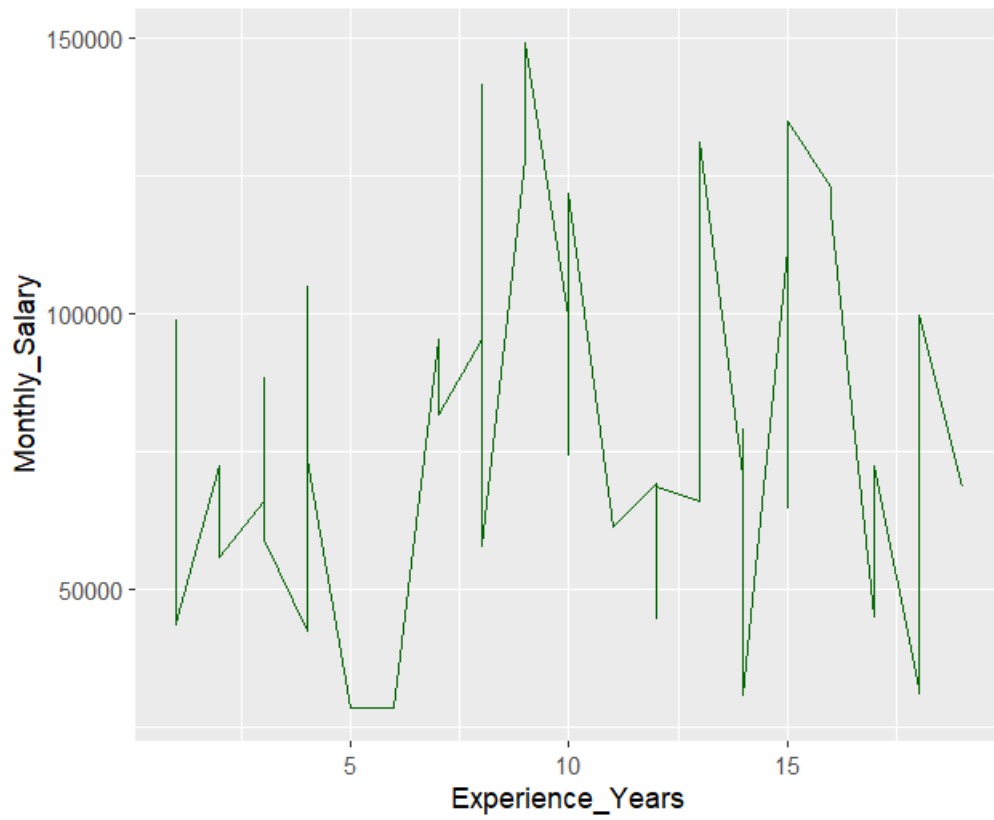


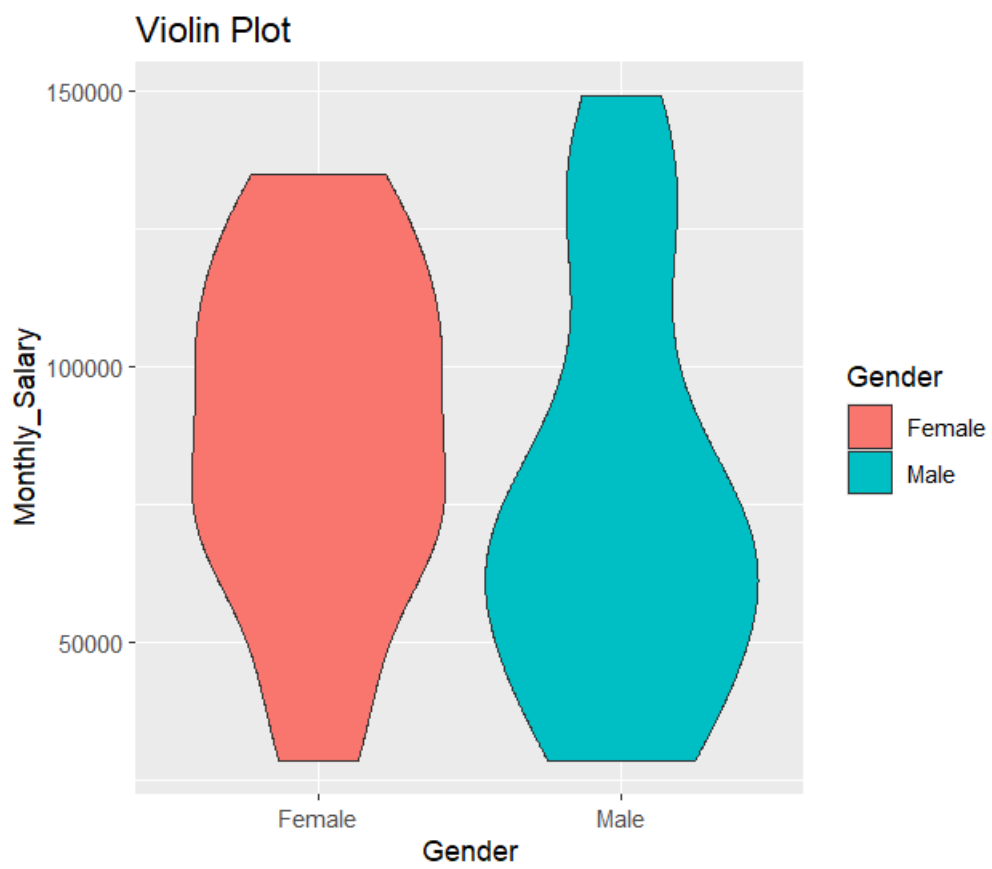
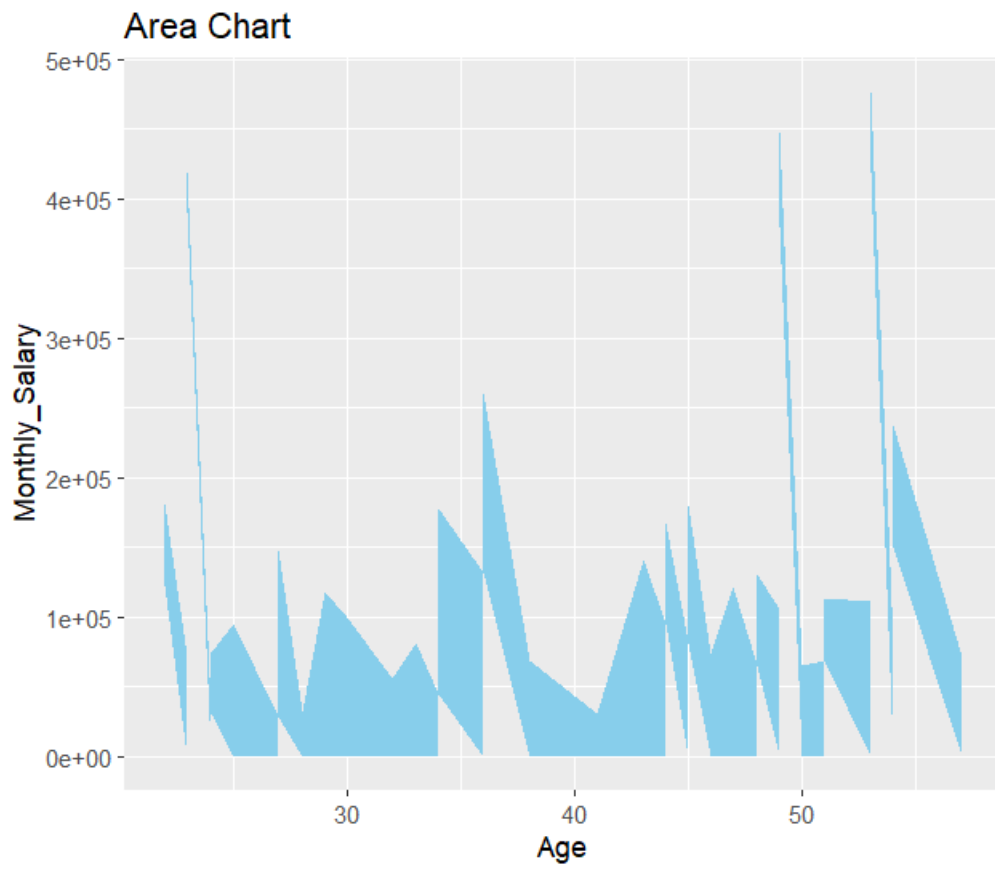


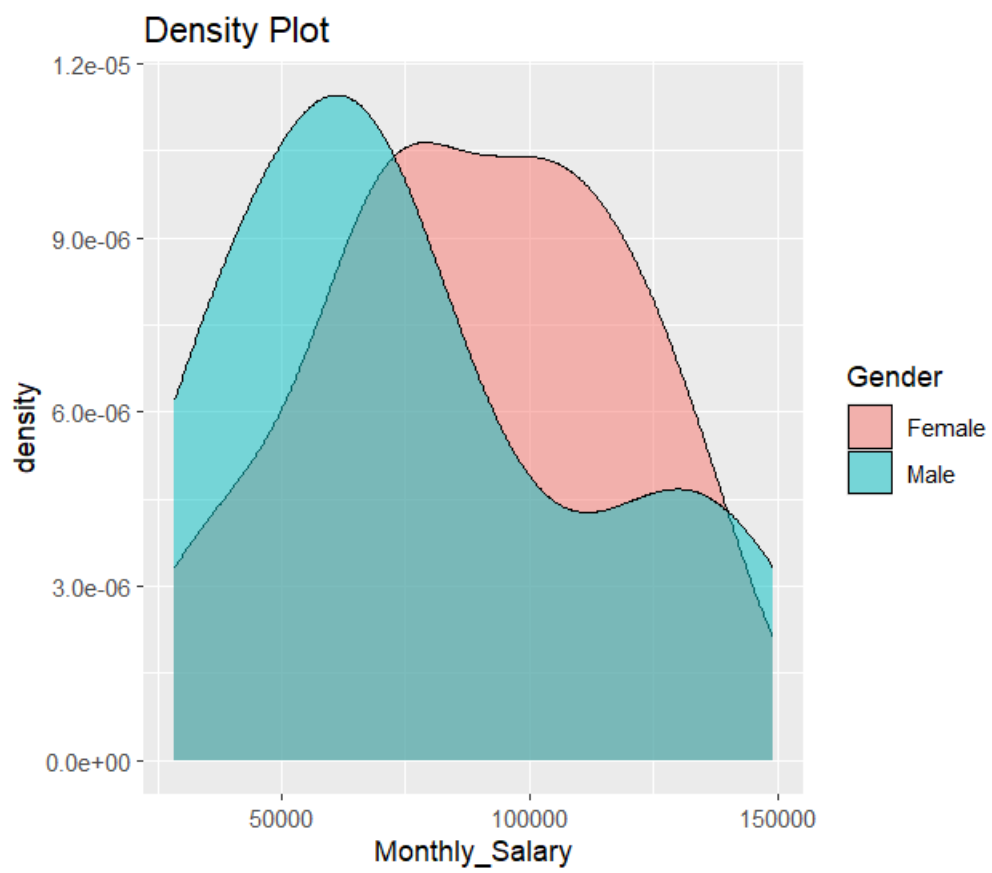
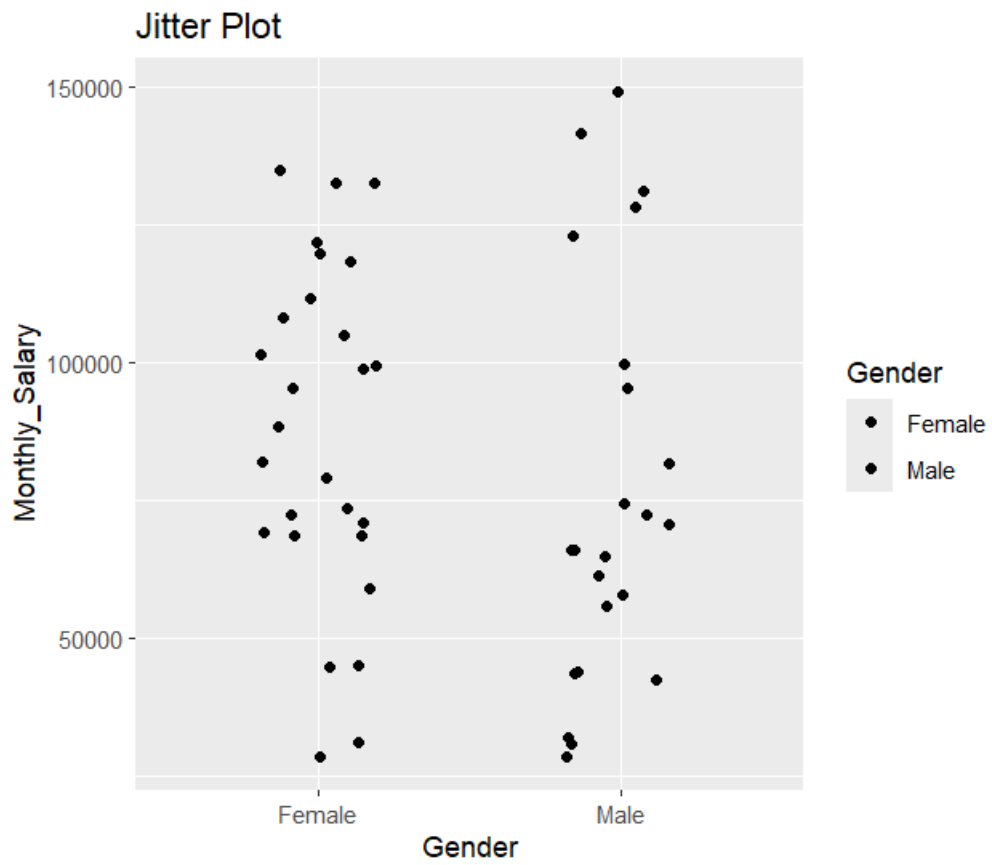
Regression Plot



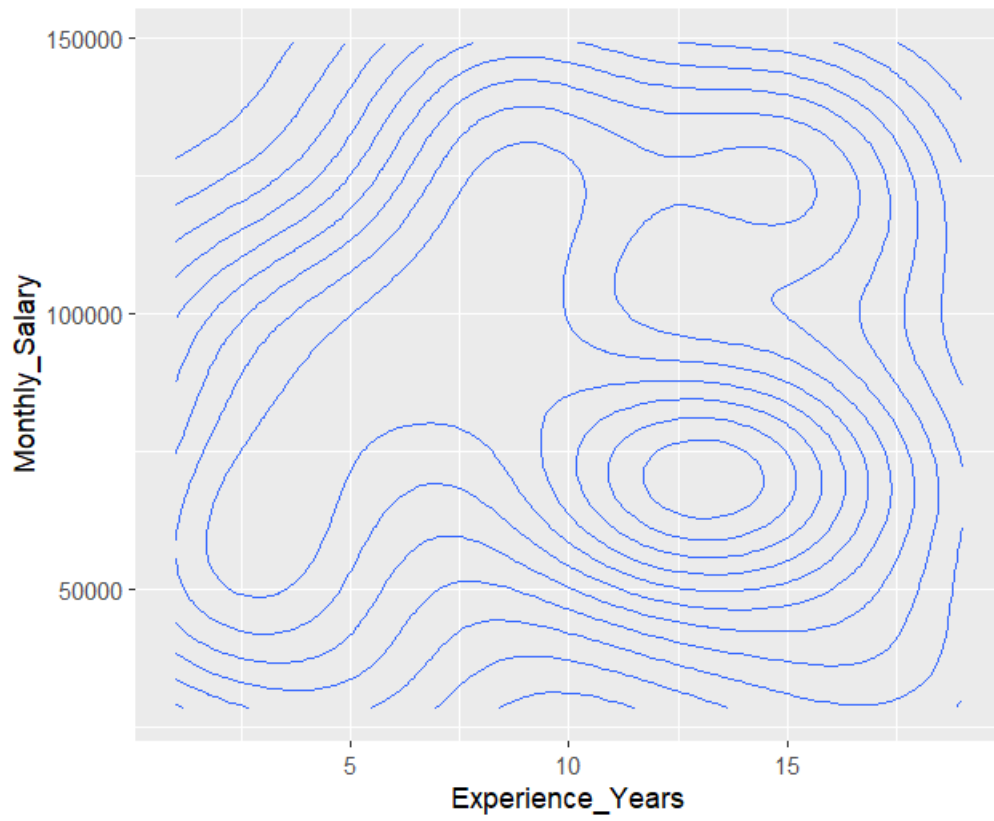
Line Chart







2D Density Plot



Contour Plot

